

SESSION - 20

Workflows



Table of Contents

Table of Contents **1**

What is a workflow in AEM? ***2***

What are workflow models? **2**

AEM Content Approval Workflow ***4***

AEM Content Approval – Workflow creation steps **4**

Exercise 1 : Creating a workflow model 5

Exercise 2 : Trigger workflow. 11

Exercise 3 : Take actions on workflow 12

Asset upload workflow overview **14**

Exercise 4: Overview and usage of DAM Update Asset workflow 14

Exercise 5: Create custom DAM Update Asset workflow 15

Exercise 6: Add a Launcher to custom DAM Update Asset workflow 18

Create custom workflow process step **21**

Exercise 7: Create your own workflow process step and use it in a workflow 21

What is a workflow in AEM?

In AEM, a workflow is a well-defined series of steps that allows users to automate the activities in the CMS instance. It consists of various steps, including people (participants), procedures (process) or some tools. The workflow can either be sequential or parallel, when multiple procedures can run parallel to each other. In short, an workflow in AEM is similar to a flowchart which describes the series of procedures to be executed by multiple participants to achieve a desired outcome.

There are many out-of-the-box workflow models readily available for use and we can even write custom workflows which can be tailored as per the business needs of the company.

There are a plethora of use cases for writing workflows depending on the business needs.

AEM Workflows allows you to automate a series of steps that are performed on (one or more) pages and/or assets.

For example, when publishing, an editor has to review the content - before a site administrator activates the page. A workflow that automates this example notifies each participant when it is time to perform their required work:

1. The author applies the workflow to the page.
2. The editor receives a work item that indicates that they are required to review the page content. When finished, they indicate that their work item is complete.
3. The site administrator then receives a work item that requests the activation of the page. When finished, they indicate that their work item is complete.

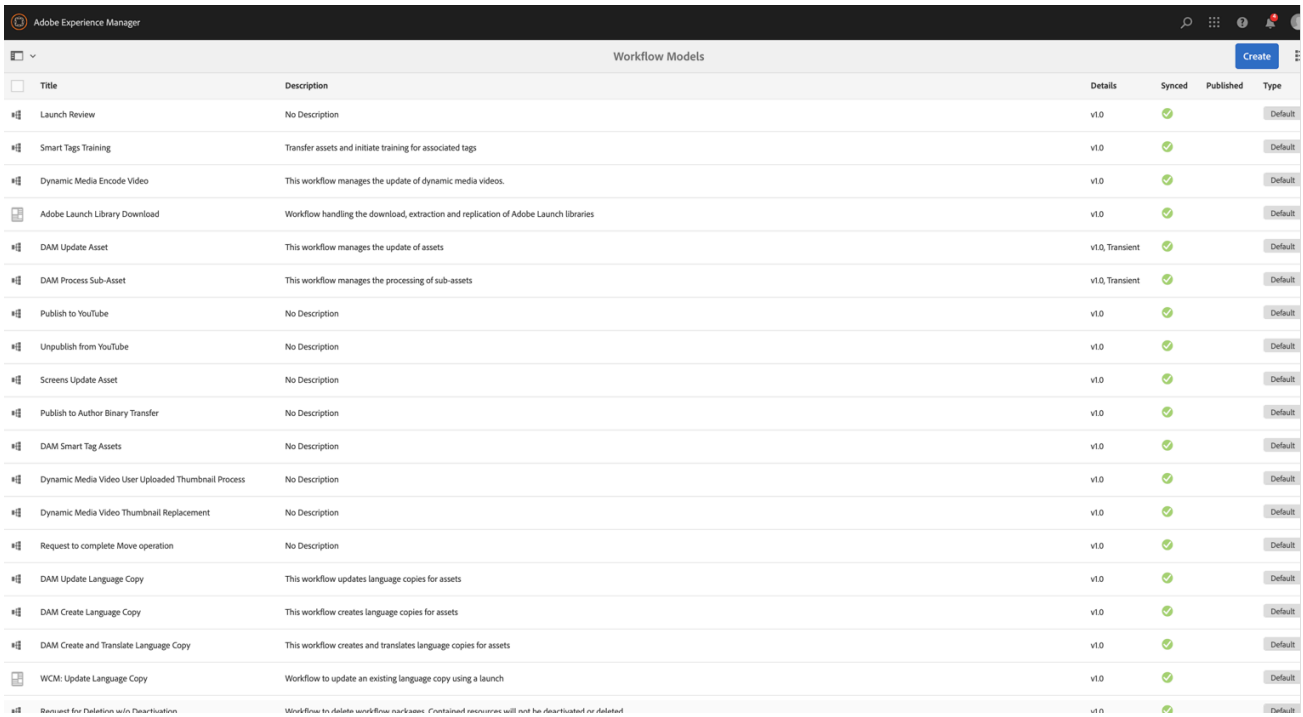
The process of publishing a page can be customized and tailored for inclusion of some special participants that need to approve the request for activation or send a mail after approval or rejection.

What are workflow models?

A workflow model is a container or blueprint of the workflow that combines the workflow steps and link them logically.

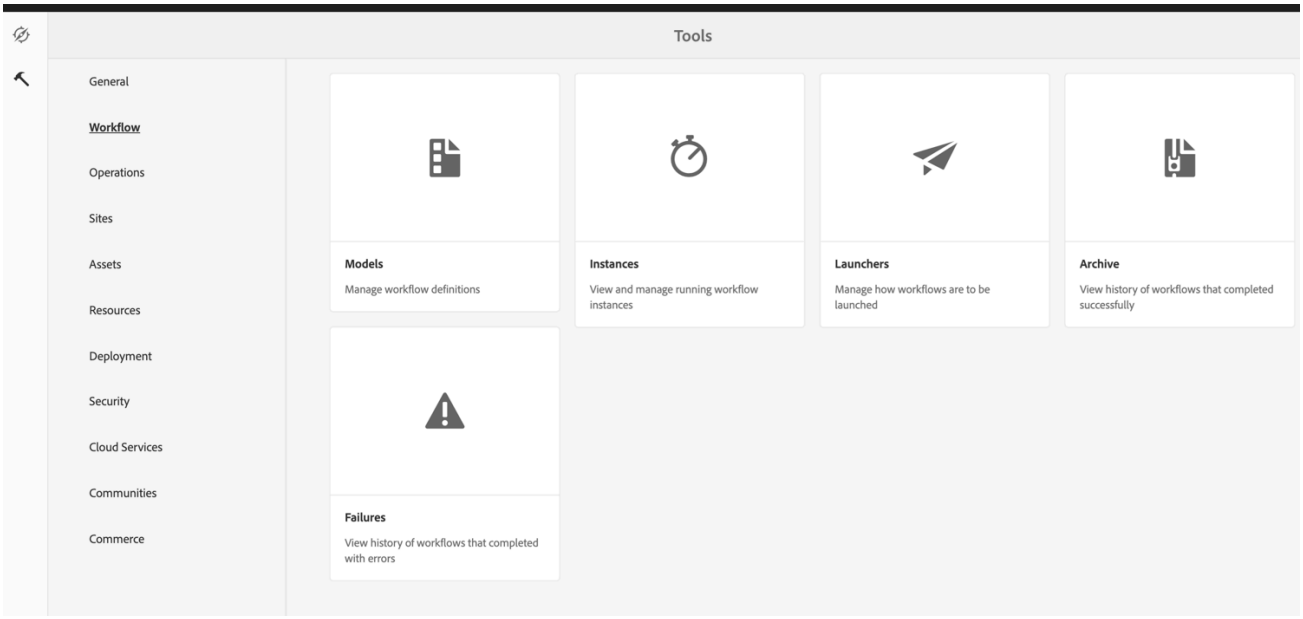
- It has a start and an end node (plus an optional participant step) by default.
- It can contain different nodes and the transitions that link them. The workflow nodes can get any input from author or can do processing via Java code or ECMA scripts.
- Workflow payload is an important term which is considered as a resource on which workflow will perform its functions.

List of workflows can be seen on
http://<host>:<port>/libs/cq/workflow/admin/console/content/models.html



These are the tabs that contain the information about the workflows.

- **Models** has workflow models having title of the workflow, version, description, and model id.
- **Instances** lists the active instances of workflows with status (running, suspended etc.), initiator, start time, workflow model, payload associated with the workflow.
- **Archive** contains the history of all the workflow instances. It contains the status (completed, aborted etc.), initiator, start and end time, workflow model, payload, workflow title etc.
- **Launcher** contains the launchers of workflow models that trigger initiation of workflows automatically on some action. It contains fields such as event type (created, deleted, modified), nodetype, globbing or path specified for action of node, condition, workflow model that needs to be triggered, description, value of enabled, exclusion list, run modes.
- **Failure** shows failed instances of workflow on which some error occurred during execution. The issues can be fixed and workflow instances can be resumed.



AEM Content Approval Workflow

There are cases when multiple department website residing on single AEM instance.

Each department needs the capability to send its content through its own approval process and specific group of people.

We needed to allow each department the capability to author its own unique approval workflow process for its authors to use and its stakeholders to participate in.

Approval workflows can easily be created out of the box with Adobe Experience Manager.



AEM Content Approval – Workflow creation steps

Steps involved in creating a workflow

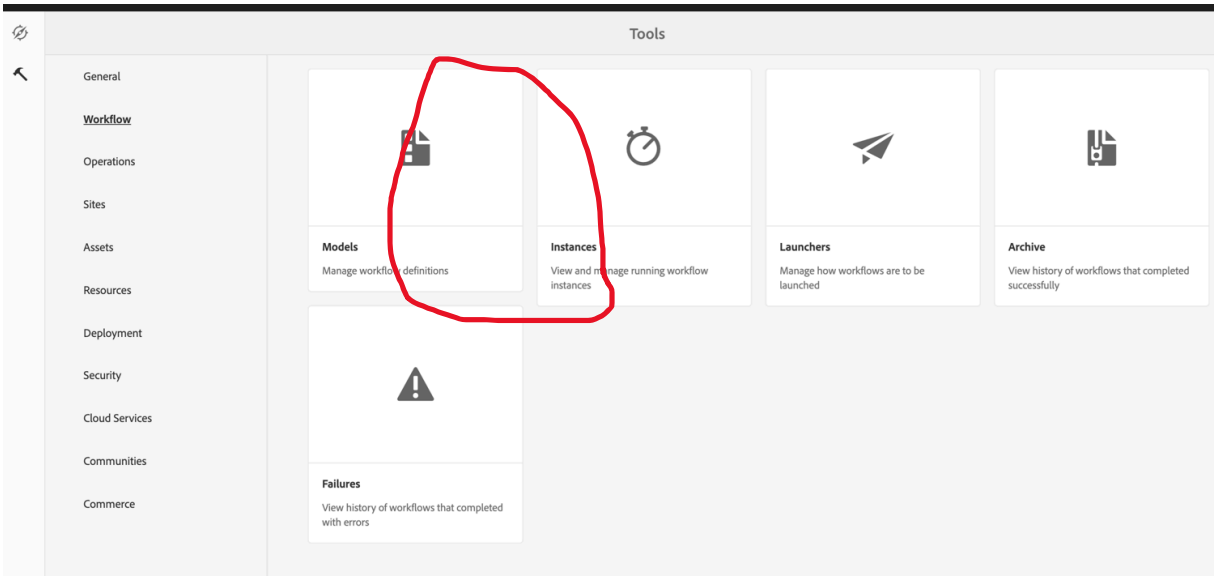
1. Create a workflow model – to define the series of steps when a user starts the workflow.
2. Trigger the workflow – 2 methods: From **workflow model** or **from page**.
3. Take actions as required so that the workflow proceeds.

4. Monitor the inbox, workflow instances section while the workflow is running.

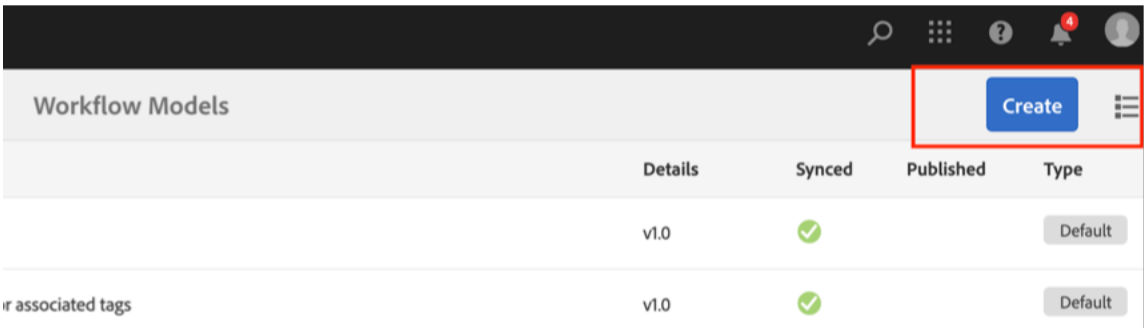
Exercise 1 : Creating a workflow model.

To create a sample AEM content approval workflow, we will first need to create a workflow model. To create a workflow model follow below steps:

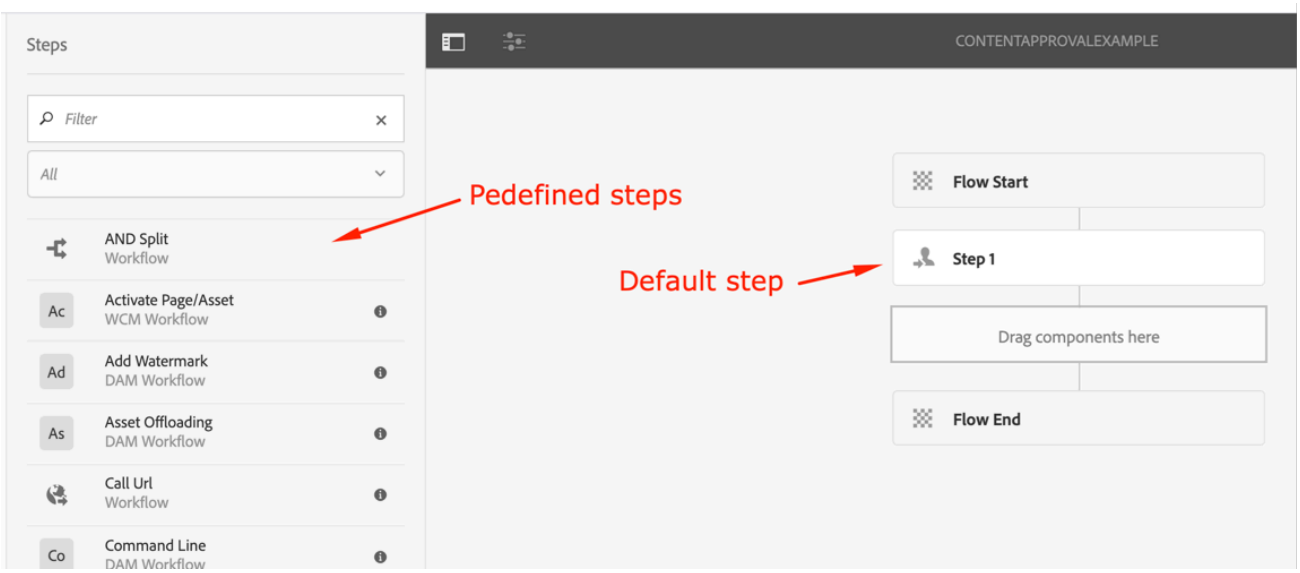
1. Go to Workflow console From Tools section as shown:



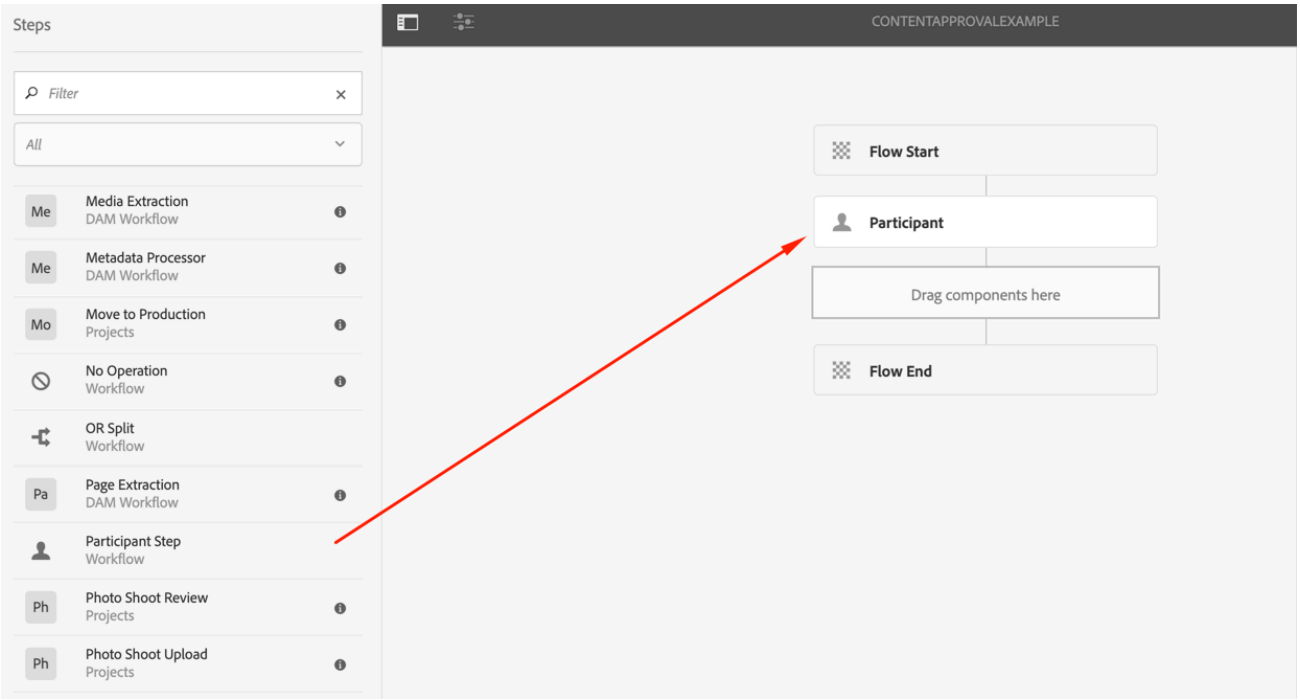
2. Now here click on create and Create Model as shown:



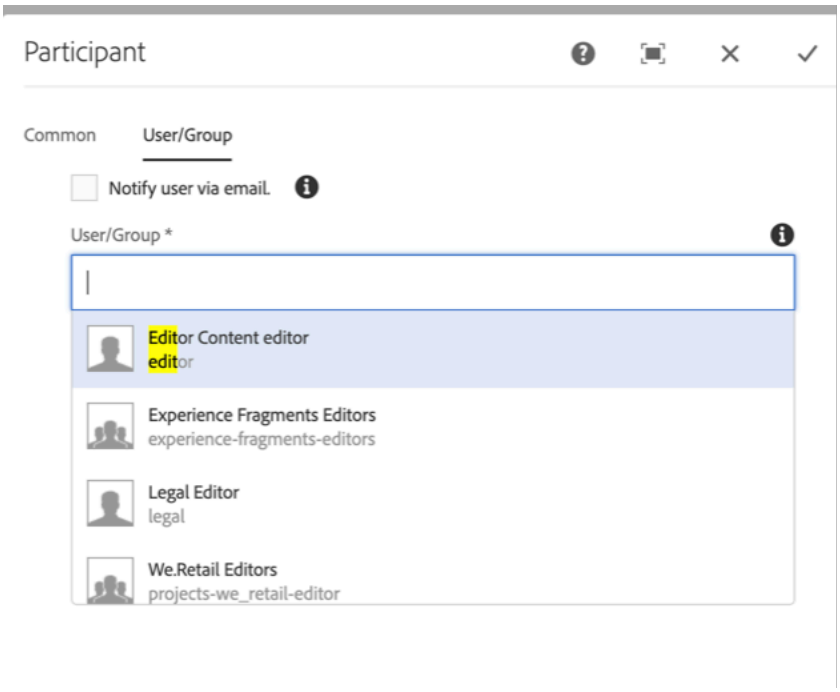
3. A dialog to enter model name will be shown. Enter then name **ContentApprovalExample** in the Title field and click on Done.
4. Now click on newly created model you will see your model with default step as below :



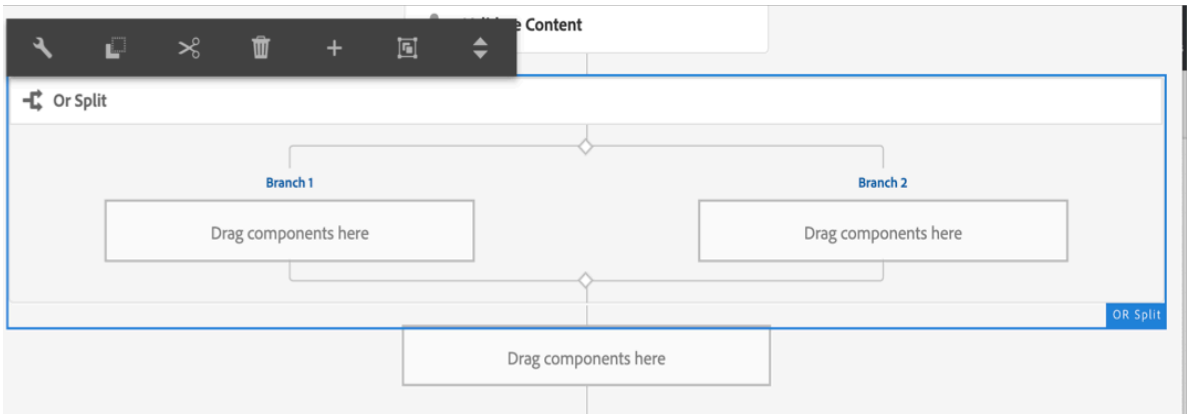
5. Now hover on the default step and **delete** it.
6. Now drag and drop a participant step into the Drag component here section.



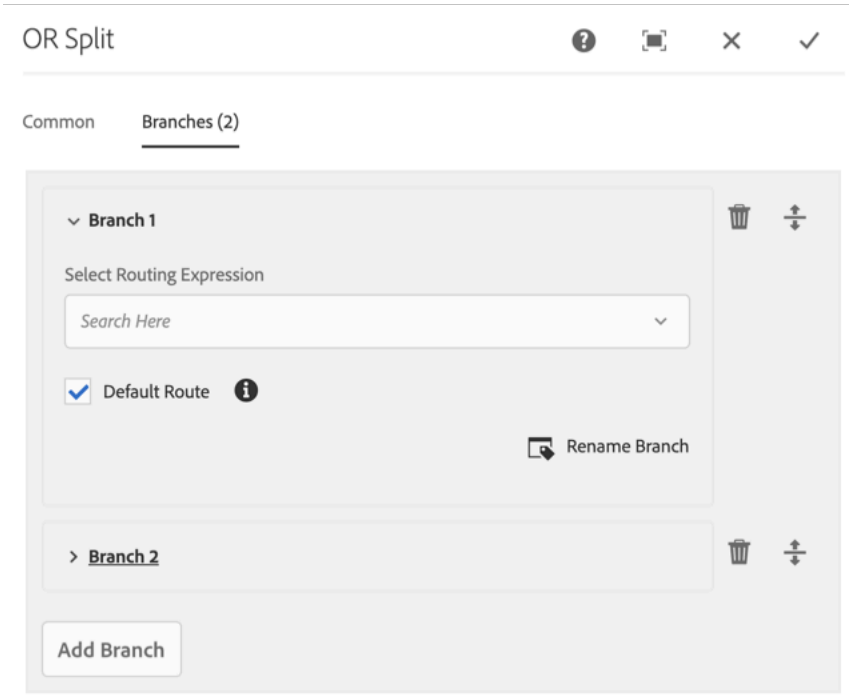
7. Configure the participant step, in the configure dialog enter title **Validate Content**.
8. In the **user/group** tab select the pre-existing user **editor** (which you created in previous exercises).



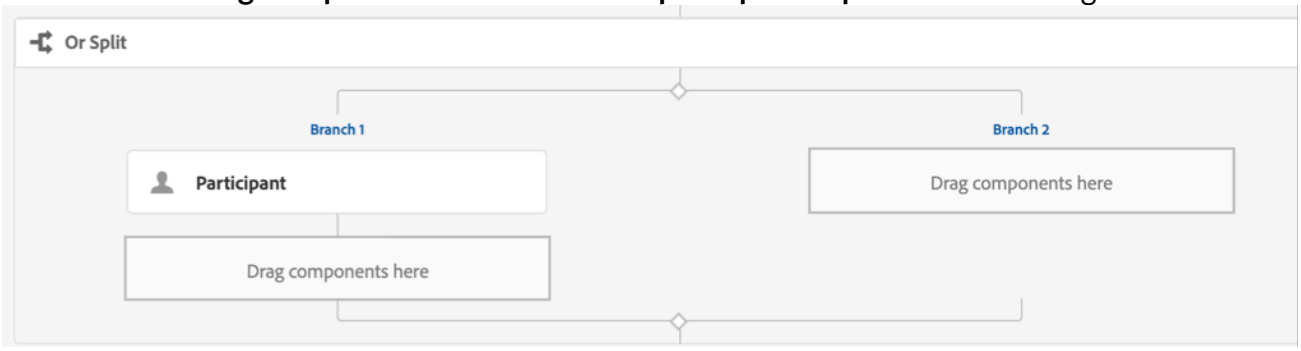
9. After this **drag and** drop an **OR split** and click on configure.



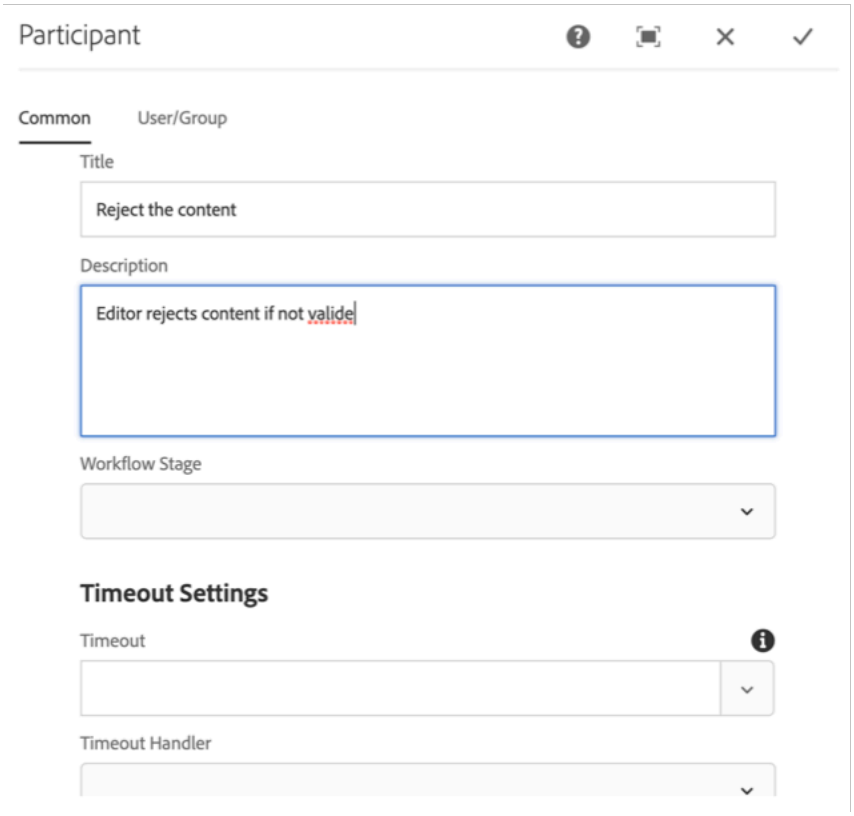
10. Select **Branch 1** as default and keep **2 branches** and leave it.



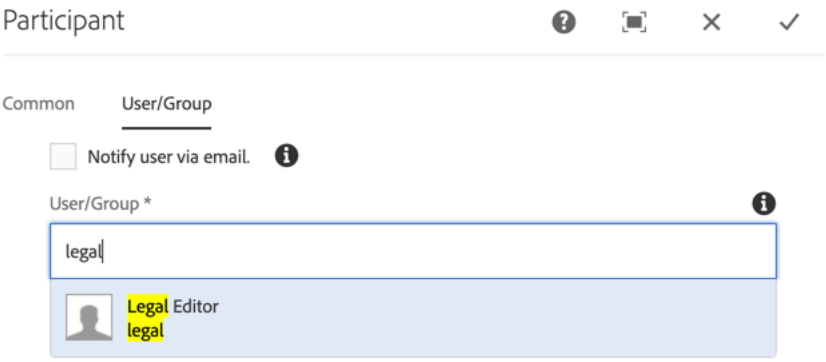
11. For first branch **drag component here** section add a **participant step** and click on configure.



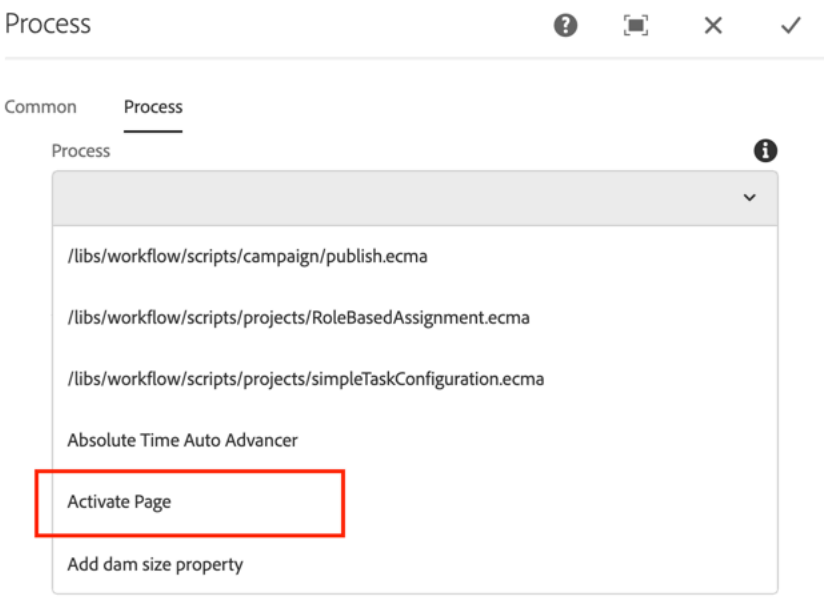
12. Rename title to **Reject the content** and add some description. Then in User/Group section add previously created user **author**.



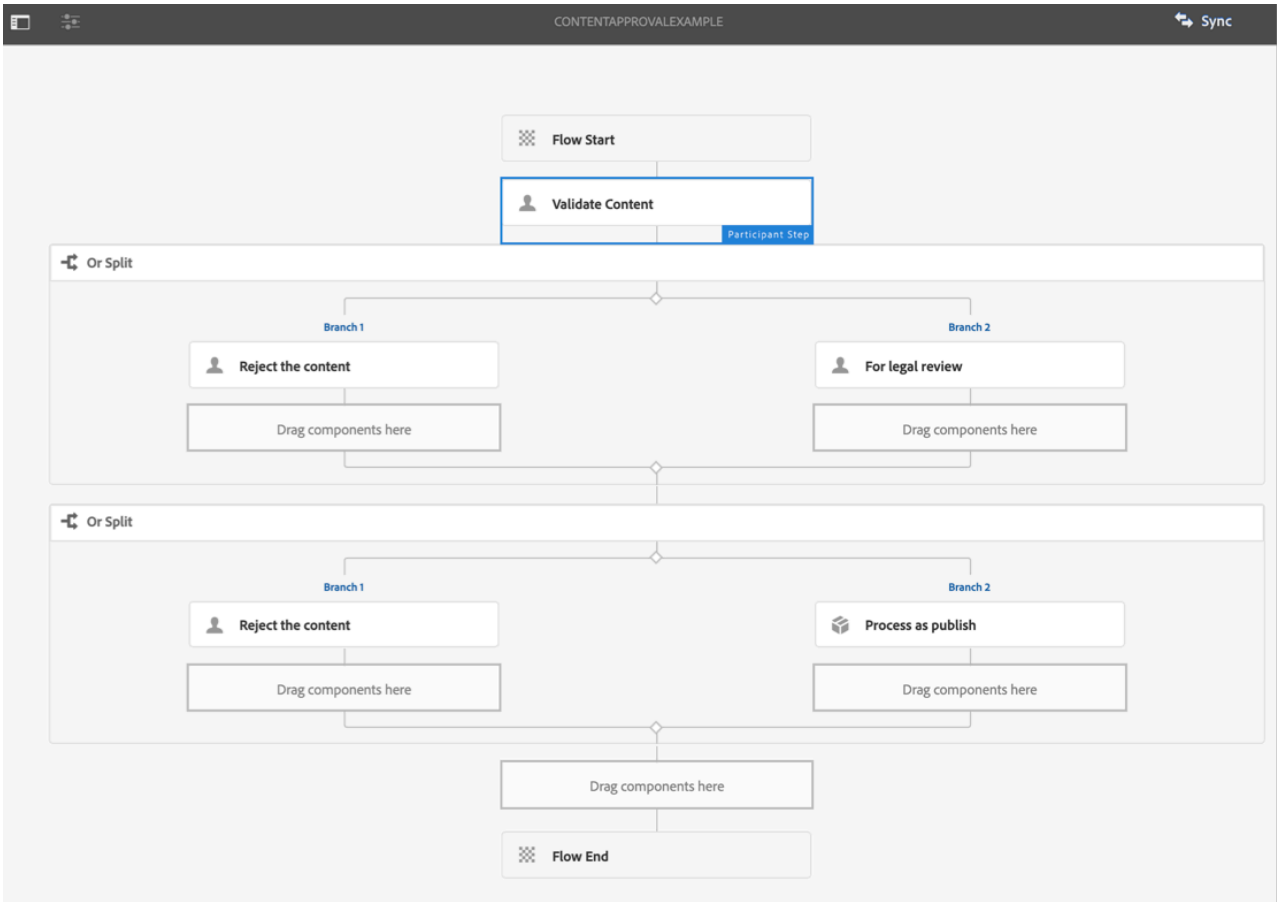
13. Similarly in **second branch** also **drag and drop** a participant step and then configure it.
14. Give title as "**For legal review**" and User/Group section add previously created user **legal**.



15. Now Add one more **OR split** as a next step. Now configure it and again **select branch 1 as default branch.**
16. Again Add a participant step in **Branch 1**, configure it and add a title "**Reject the content**".
17. Now in **User/Group section** add user **author**, so that flow go back to author for next step.
18. Now add a **process step** in **branch 2** and click on configure it. Give title as "**Publish as requested**".
19. In the Process tab select Process "Activate Page". Also select the **Handler Advance** check box.



20. Now your **Workflow Model** is completed and looks like below :



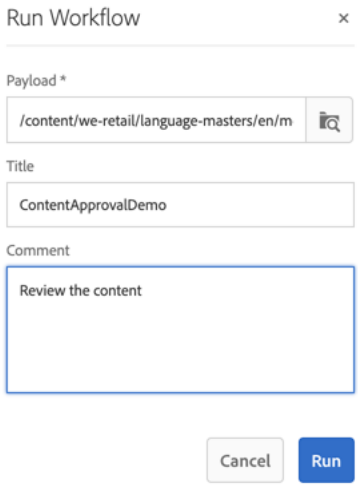
21. Once Model is created **click on Sync button** at the top right hand side.

Exercise 2 : Trigger workflow.

We can trigger a workflow either of the below two methods
Method 1 : Trigger from workflow console and going into Models sections.
Method 2 : Trigger from the actual page.

Triggering a workflow from workflow console -:

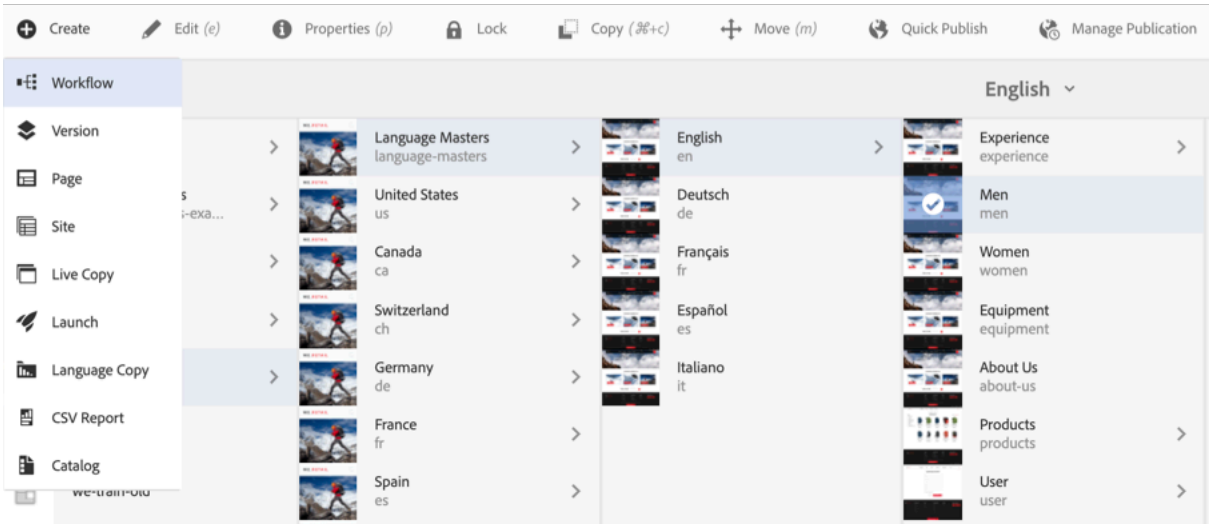
1. **Login to AEM as author user** which you created earlier.
2. Go to Workflow console From Tools section and go to Models:
3. Select your newly created model and click on **Start Workflow** button in the top menu.
4. You will see a dialog like this. here select the content/page which you want to proceed for approval process as a payload. This **payload** is the input for workflow. Also provide the Title (**ContentApprovalDemo**)and description (**Review the content**) for the workflow.



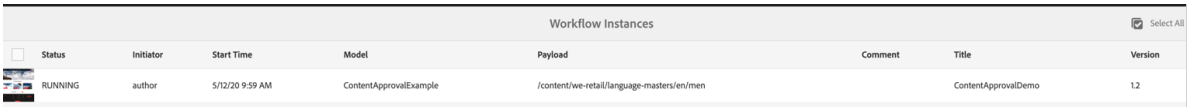
5. Now click on Run. Your workflow has been started.

Triggering a workflow from page -:

1. **Login to AEM as author user** which you created earlier.
2. Go to sites console and select a page. Now click on create button from the top menu and click on **Workflow**.



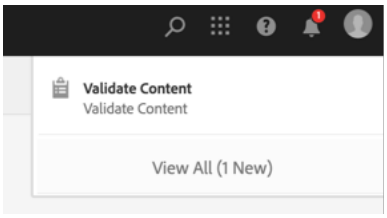
3. Now on the dialog select the workflow model (**ContentApprovalExample**) and provide title provide the Title (**ContentApprovalDemo**).
4. Now on next screen select the page as payload and click on create. You have triggered the workflow for content approval using the model **ContentApprovalExample**.
5. To verify go to workflow console and click on instances. You will see your workflow as Running status as shown:



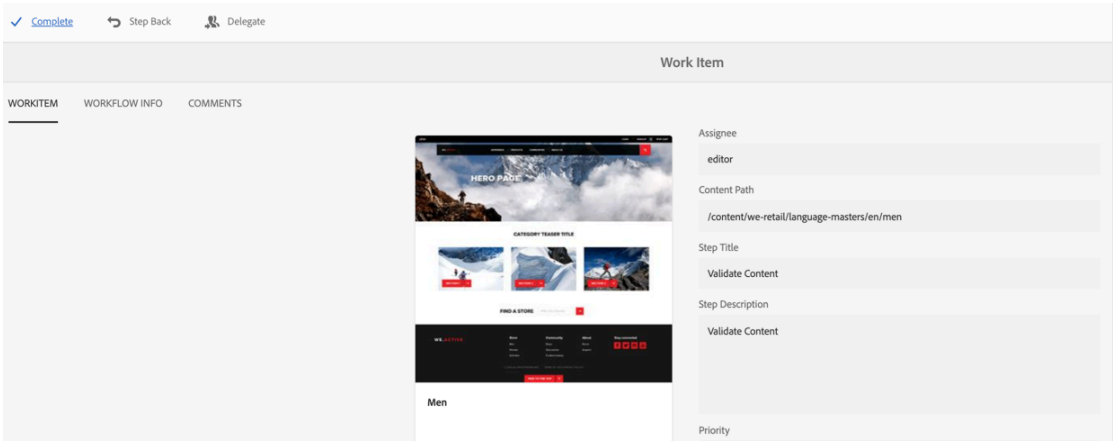
Exercise 3 : Take actions on workflow

Since you were logged in as author, you will not be able to see the next action items for this workflow until editor logs in and proceed for the next step.

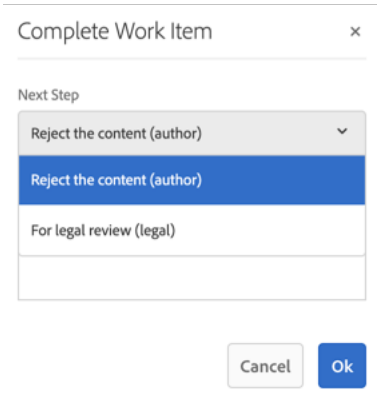
1. So now logout as author and then login as editor.
2. You will see a notification click on it.



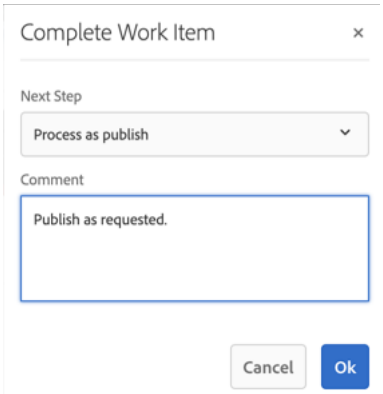
3. Clicking on the notification you will see that you got an action item for you to perform in AEM inbox as shown.



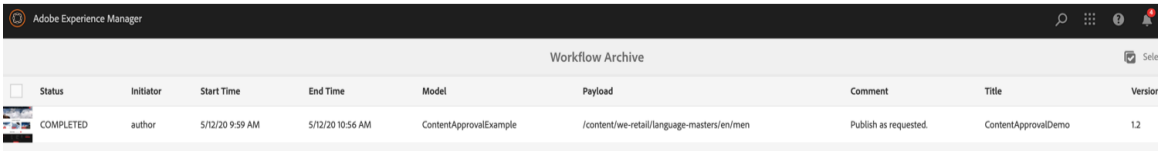
4. Before proceeding further go and check in workflow console Instances. There you will not see anything because for editor user workflow still in inbox and not started.
5. Now click on Complete the button and you will see two options in the dropdown as we have provided using OR split.



6. Select the option **For legal review**. Also provide your comment.
7. Now it has gone for **approval to legal person**. So logout and login as **legal user.**
8. After login go to AEM inbox via clicking on notification. There you will see **workItem** in your inbox. Click on **Complete** and select the "**Process as Publish**" to approve it.



9. Now your workflow has been completed. To check this login as admin and see in the archive section from workflow console. You will see your completed workflow list as below:



Asset upload workflow overview

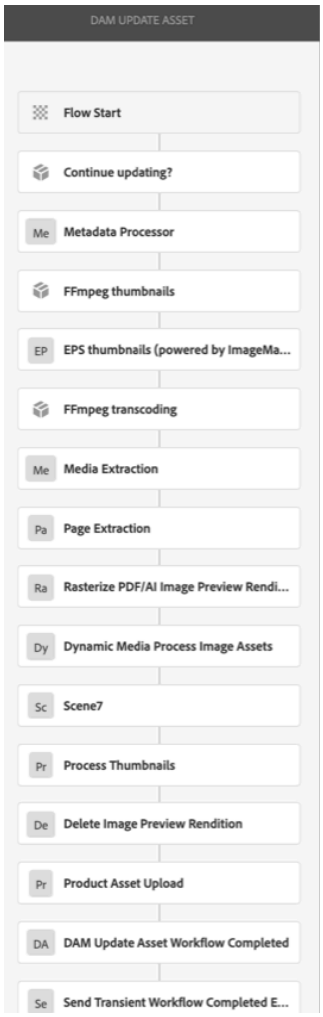
DAM Update Asset Worfklow -: DAM Update asset is a default workflow which is triggered whenever any asset is upladed into the DAM. This workflow helps to limit the size and the quality of the image displayed.

Some of the usage of asset workflows

1. Create image renditions.
    - Using different size images on the websites
    - Images for different devices or viewport.
2. Updating image metadata.

Exercise 4: Overview and usage of DAM Update Asset workflow

Go to workflow console and go inside Models section. There from the list of models select the DAM Update Asset workflow and click on edit. You will see list of steps in the workflow model as shown:



Note : DAM Update Asset workflow is a Launcher workflow, that means whenever any asset is uploaded into DAM this workflow gets triggered.
Note : To create custom renditions we can create our own customize workflow.

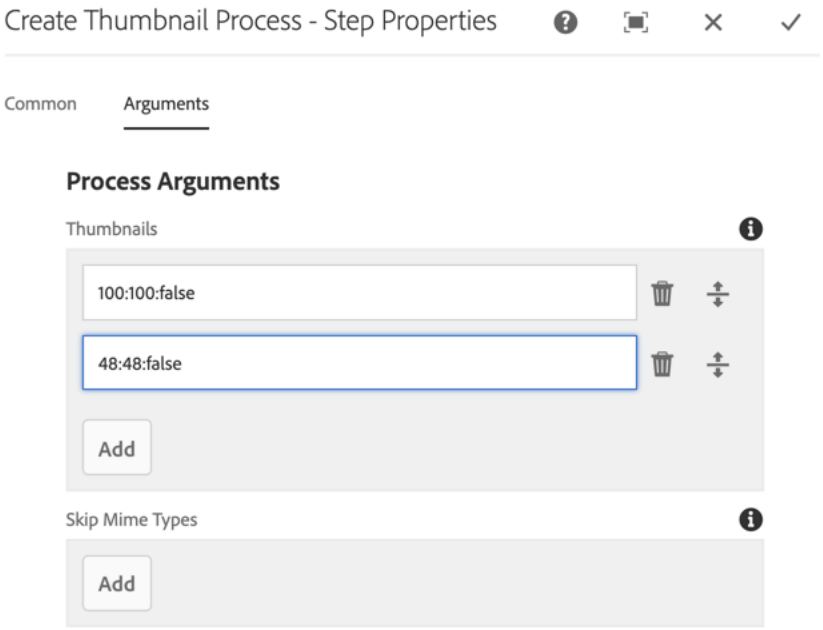
1. Understand the various steps from the DAM Update Asset workflow.
2. Upload an image into AEM and see
   - the renditions created
   - metadata updated

Exercise 5: Create custom DAM Update Asset workflow

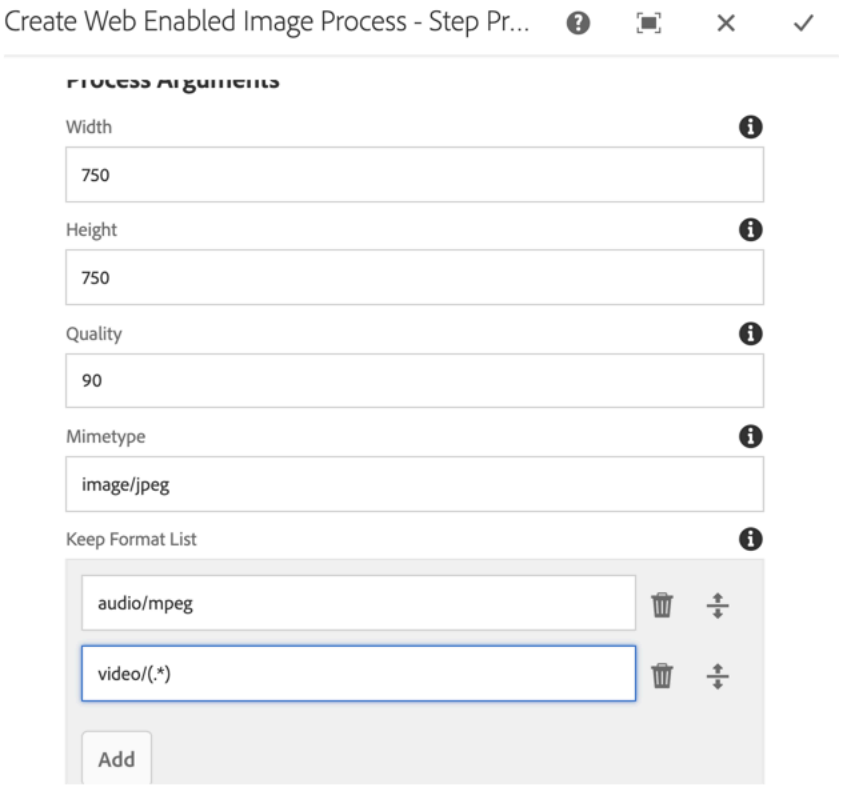
We will create our own custom DAM update Asset workflow. This workflow will be t**riggered manually**
To do that follow below steps

1. Create a new Workflow model. Give the title "**My manual trigger workflow**".
2. Go to workflow console and go to Models section. In Models section go and edit this newly created Models.

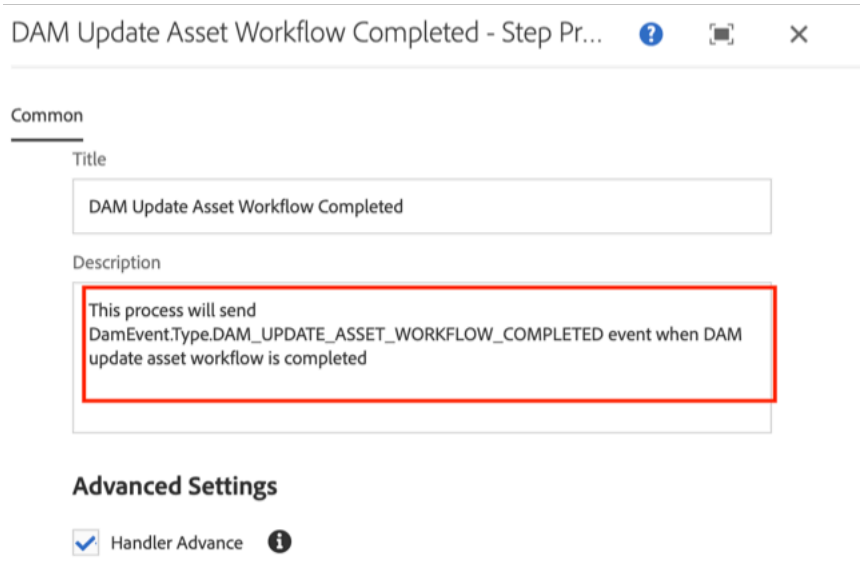
3. Here in the Model delete the default step 1 and add a **Create Thumbnail** step.
4. Configure this step and add two thumbnail option as shown in the image below :



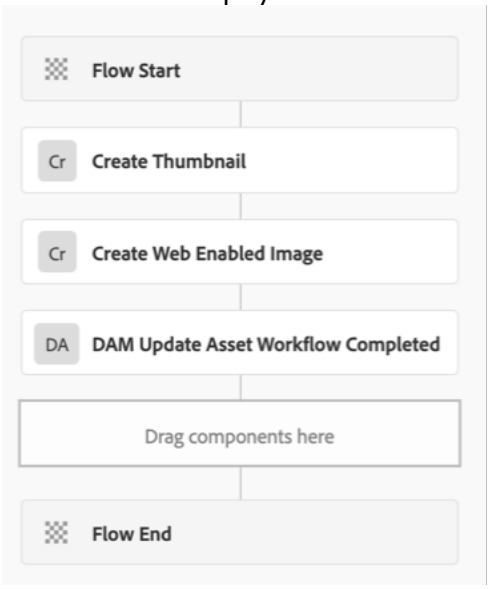
5. Now add a new step Create Web Enabled Image and configure it like the below image for process arguments tab only.



6. Now add a step **Dam Update Asset Wokflow Completed** step and click on configure.
7. In the dialog tab remove the description text as shown in the below image.



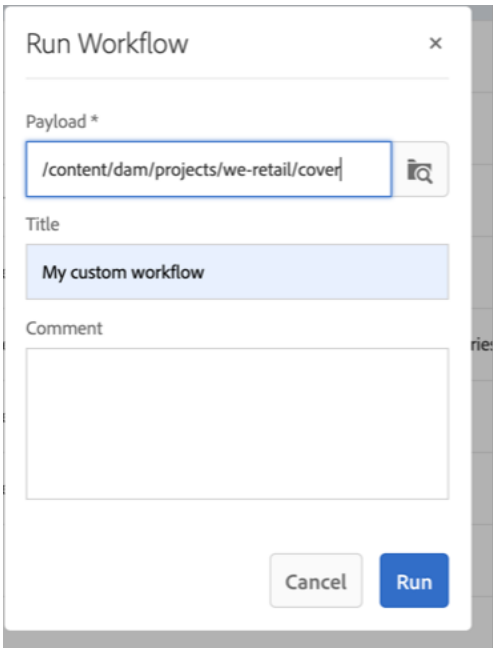
8. After all above steps your model should look like this :



9. Now your model is ready click on **sync** button.

Now we will trigger this custom workflow manually.
To do that follow below steps :

10. Select the workflow model and click on start workflow button from top menu.
11. In the opened dialog select any image from the DAM in payload section and give title for the workflow as shown :



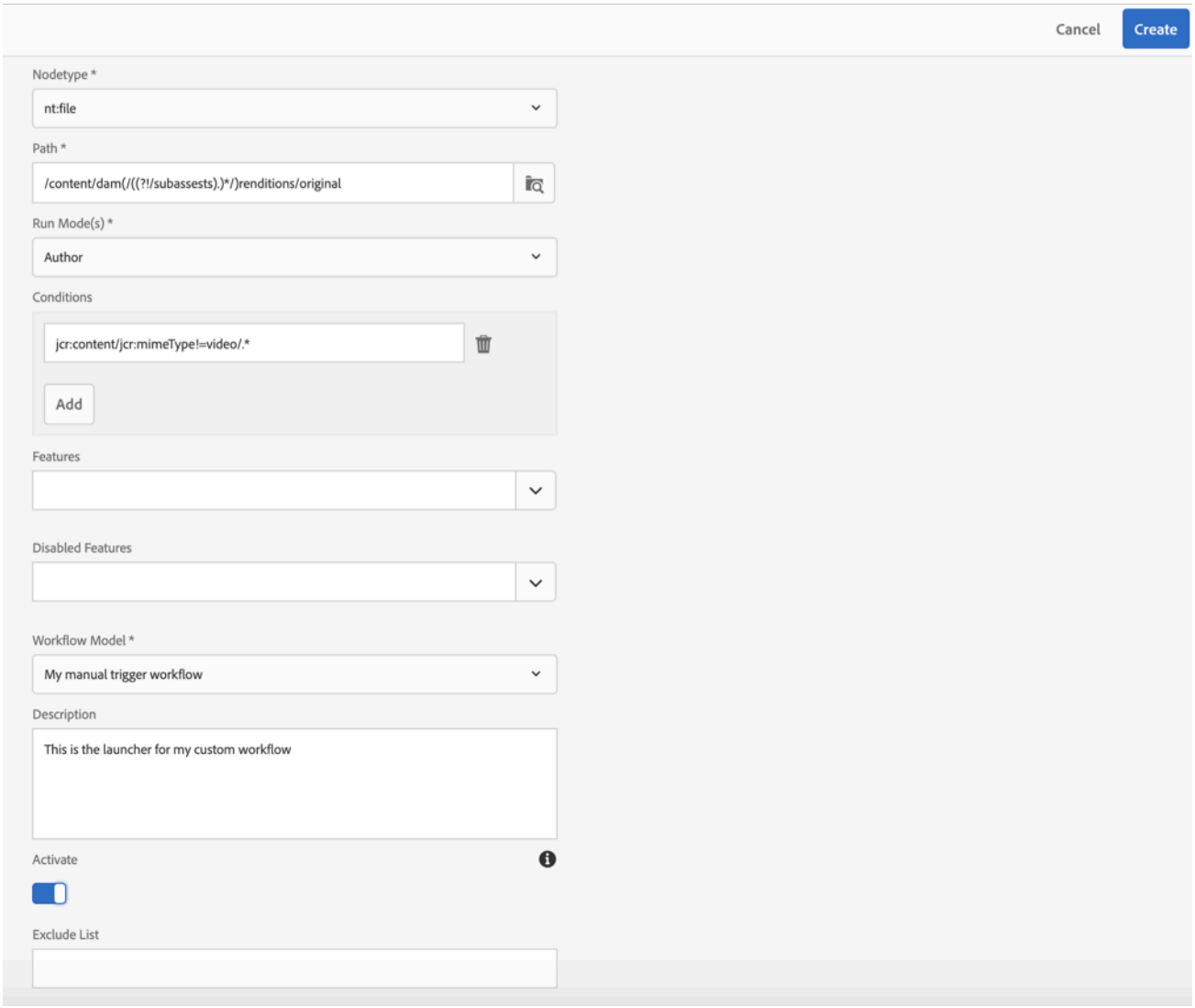
12. Now click on run. You can see your workflow ran and see it in the archive tab in workflow console.

Exercise 6: Add a Launcher to custom DAM Update Asset workflow

In previous execise we have created a custom workflow and triggered it manually. Now we will see how we can trigger a workflow automaticallly from a launcher. Let us see how to create a launcher workflow from the following steps :

1. Go to **workflow console** and click on **launcher tab**.
2. Here in the launcher tab click on create button and click on **Add Launcher**.
3. A configuration window will open from which we will create a launcher for the workflow.
4. Enter the values as
   Event Type : Created
   Nodetype : nt:file
   Path -: /content/dam(/((?!/subassests).)*/)renditions/original
   Run Modes : author
   Conditions : jcr:content/jcr:mimeType!=video/.*

   Wokflow Model -: Select your custom workflow created in previous section.
5. Then add **description** for the launcher as well.
6. Now click on activate button and then click on Create button. Your launcher is created as shown from below image:



7. You can see it in the launcher list.

To test this launcher workflow getting triggered, we need to disabled OOTB dam update asset workflow launcher other wise we will not be able to test our custom launcher. To do that.

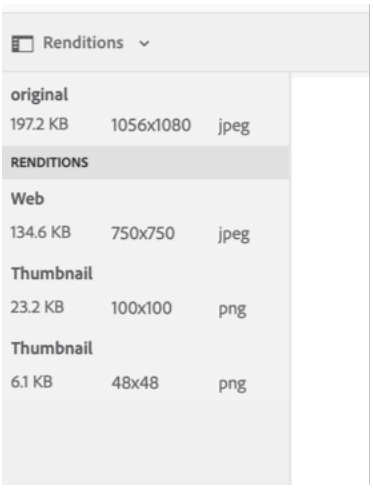
8. Search for "Update Asset" text and select the launcher and click on properties as shown.

Properties | Quick Publish | Manage Publication | Delete

Workflow Launchers

Globbing	Description	Event Type	Nodetype	Condition
/content/dam(/.*/)jcr:content	Dynamic Media Video Thumbnail Replacement	Node modified	nt:unstructured	dam:thumbnailProxyUrl!=
/content/dam(/.*/)metadata		Node modified	nt:unstructured	dc:format!=
/content/forms/fp	Replicate all the drafts/submissions that are just modified	Node modified	nt:unstructured	
/content/dam		Node modified	sling:Folder	jcr:content/autoTag==true
/home/users/screens(/.*/)devices(/.*/)profile_screens(/.*)	Replicate binary data from pubish to author for screen devices's content	Node created	nt:file	
/content/dam(/.*/)(subassets)(/.*/)renditions/original	Process Subasset	Node created	nt:file	jcr:content/jcr:mimeType!=video/.*
/content/dam(/((?!/subassests).)*/)renditions/original	This is the launcher for my custom workflow	Node created	nt:file	jcr:content/jcr:mimeType!=video/.*
/content/dam(/((?!/subassets).)*/)renditions/original	Update Asset - Create	Node created	nt:file	jcr:content/jcr:mimeType!=video/.*
/content/dam(/.*/)renditions/original	Parse Word documents (DOC) that have been added to the DAM	Node created	nt:file	jcr:content/jcr:mimeType==application/msv
/content/dam(/.*/)renditions/original	Dynamic Media Encode Video - Create (post-6.3)	Node created	nt:file	jcr:content/jcr:mimeType==video/.*
/content/dam(/.*/)manualThumbnail.jpg	Dynamic Media Video User Uploaded Thumbnail Process	Node created	nt:file	../dam:s7damType!=
/content/dam(/.*/)renditions/original	Parse Word documents (DOCX) that have been added to the DAM	Node created	nt:file	jcr:content/jcr:mimeType==application/vnd officedocument.wordprocessingml.docume
/content/dam(/.*/)renditions/original	Dynamic Media Encode Video - Create	Node created	nt:file	jcr:content/jcr:mimeType==video/.*
/content/dam(/.*/)renditions/original	Update Asset - Create (with DM feature check)	Node created	nt:file	jcr:content/jcr:mimeType==video/.*

9. From the properties window, deactivate the launcher and save it.
10. Do the same for any similar DAM Update launchers.
11. Now we will test this launcher via uploading an image to content dam.
12. Before doing this verify that what is the last ran workflow from archive tab of workflow console.
13. Now upload an image. After uploading the image you can verify the renditions which we have added into our custom workflow.



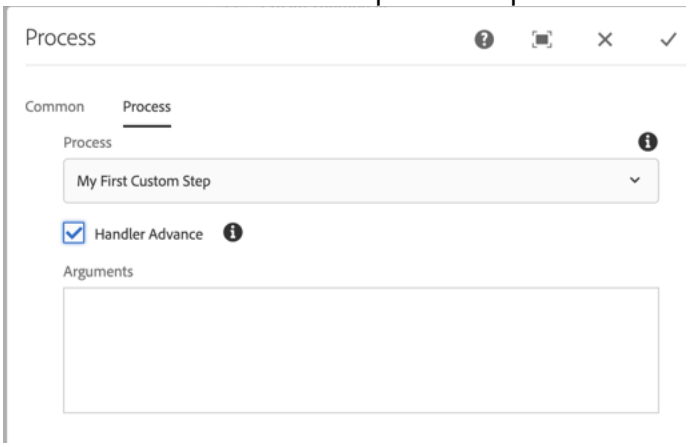
14. Also you can see it in the archive tab of the workflow console.

Create custom workflow process step

Till now we have seen out of the box workflow steps. In this exercise we will create our own workflow process step. To do that we need to create our own java class with specific annotations. Let'ss do this in the following exercise.

Exercise 7: Create your own workflow process step and use it in a workflow

1. Create a new package with the name workflows..
2. Now in this package create a new class with the name "**CustomWorkflowProcess.java**"
3. Here in this class you can copy the code from the exercise folder.
4. Now build your code and deploy it on server.
5. Now create a new workflow model
6. In that workflow model add a process step and select our custom process step which we created.



7. Also don't forget to select **Handler Advance** check box to proceed it automatically for next step.
8. Now sync this model.
9. After this you can start this workflow and verify the logs in the log file.